

Data Analyst Professional Practical Exam Submission

You can use any tool that you want to do your analysis and create visualizations. Use this template to write up your summary for submission.

You can use any markdown formatting you wish. If you are not familiar with Markdown, read the [Markdown Guide](#) before you start.

Task List

Your written report should include written text summaries and graphics of the following:

- Data validation:
 - Describe validation and cleaning steps for every column in the data
- Exploratory Analysis:
 - Include two different graphics showing single variables only to demonstrate the characteristics of data
 - Include at least one graphic showing two or more variables to represent the relationship between features
 - Describe your findings
- Definition of a metric for the business to monitor
 - How should the business use the metric to monitor the business problem
 - Can you estimate initial value(s) for the metric based on the current data
- Final summary including recommendations that the business should undertake

Start writing report here..

Executive Summary

This report addresses the need to improve the sales strategy at Pens and Printers by analyzing customer data derived from three communication methods: email, call, and a combination of both. The analysis aims to identify customer engagement effectiveness, revenue distribution, and trends over time, ultimately providing recommendations for the most efficient and profitable sales approach. The insights derived from this report will facilitate better business decisions and optimize customer engagement strategies for the company.

Business Problem

Pens and Printers is seeking to enhance its sales strategy through an in-depth analysis of customer engagement methods. Key questions guiding this analysis include:

- How many customers engage through each communication method?
- What is the revenue distribution across different methods?
- How has revenue changed over time for each method?
- Which method demonstrates the highest profitability and efficiency?
- Are there any significant differences among customer groups based on their communication method?

The findings from this analysis will empower Pens and Printers to make informed decisions about future sales strategies.

Data Overview

The dataset, sourced from the company's sales tracking system, comprises 15,000 rows and 8 columns, including:

- week
- sales_method
- customer_id
- nb_sold (number sold)
- revenue
- years_as_customer
- nb_site_visits (number of visits on website)
- state

The data meets the ROCCC criteria—Reliable, Objective, Comprehensive, Current, and Consistent—indicating its credibility for analysis. Initial data integrity checks revealed missing values in the 'revenue' column, which were addressed by excluding these rows due to their limited impact. Further cleaning involved converting data types, consolidating variations in the 'sales_method' column, and confirming the consistency of customer tenure data.

Data Cleaning and Preparation

- **Missing Values:** Over a thousand rows with missing 'revenue' values were dropped.
- **Duplicates:** A thorough check for duplicate entries confirmed that there were no duplicates within the dataset. Additionally, we verified that no customers were repeated in the 'customer_id' column.
- **Data Types:** Converted 'sales_method' and 'state' columns to string types.
- **Consolidation:** Merged variations in the 'sales_method' column to reduce categories from five to three.
- **Referential Integrity:** The 'years_as_customer' column was examined to ensure no values fell below 40 years, which is consistent with the founding year of Pen and Printers (1984). All 50 states were represented in the 'state' column without errors.
- **Statistical Summaries:** Preliminary statistical analysis indicated that the average number of new products sold was 10, with a revenue range between 32.50 and 238.30, averaging \$93.90. Customer tenure averaged five years, with a notable standard deviation.
- **Outlier Examination:** After addressing outliers in the 'years_as_customer' column, we found no significant outliers in the 'revenue' or 'nb_sold' columns.
- **New Columns:** Added 'revenue_per_item' (revenue/nb_sold) and 'customer_category' for enhanced analysis.
- The dataset is now saved as a new .csv file, ready for further aggregation and analysis.

Insights from Customer Sales Methods

The analysis reveals distinct trends in customer engagement based on the communication method:

- **Customer Distribution:**
 - o Email: 6,921 customers (50% of total)
 - o Call: 4,780 customers (34%)
 - o Email + Call: 2,223 customers (16%)

Revenue Analysis

- o The most profitable week recorded \$272,810.
- o Sales performance fluctuated significantly, with the lowest revenue of \$163,111 occurring in the 6th week.
- Analyzing the data reveals that the Email method is the primary sales approach, accounting for 6,921 customers, which is approximately 50% of the total customer base. The Call method follows, with 4,780 customers, representing 34% of the total. Lastly, the combined Email + Call method consists of 2,223 customers, making up 16% of the total.

Revenue by Sales Method

- Email: Generated \$672,220 (53% of total revenue).
- Email + Call: Contributed \$408,256 (31%).
- Call: Accounted for \$227,513 (17%).
- Trends show that the Email method peaked early but experienced a decline, while the Email + Call method gained momentum, surpassing the Email method in revenue by the 4th week.
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Key Observations on Sales Performance

The Email method consistently generated the highest revenue, with its best performance yielding \$229,765, while the Call method lagged significantly with a peak of only \$53,518. The Email method's efficiency, in terms of revenue generation per effort, makes it the clear leader, whereas the Call method's low contribution suggests it should be deprioritized in future strategies.

Trends and Patterns in Customer Behavior

An exploratory analysis of customer behavior revealed significant differences in revenue contribution across various states. California ranked highest in revenue, generating over \$160,893 in six weeks, followed by Texas and New York. This trend aligns with the demographic and economic characteristics of these states, which have large populations and robust economies. Conversely, states with the lowest purchases, such as Maine and North Dakota, showed revenue contributions ranging between \$2,600 and \$5,300.

Furthermore, a breakdown of customer revenue contributions over time indicates that the five years account for more than 60% of total sales, amounting to. After the fifth year, revenue contributions decrease significantly (34%), with customers after ten years contributing minimally (14%).

Additionally, the Email method drove significant website visits (over 171,000), outperforming the Call method (over 116,000) and the combined method (60,000). This underscores the Email method's effectiveness not only in revenue generation but also in customer engagement.

Metrics

To ensure continuous monitoring of the company's sales strategy and measure success in addressing the current challenges, we propose using **Revenue per Customer by Communication Method (RPC)** as a key performance metric. This metric tracks the average revenue generated per customer for each communication method—Email, Call, and Email + Call—providing a clear indication of the profitability and engagement efficiency of each approach. Using existing data as a baseline, the RPC for the Email method is calculated at \$97.12 per customer, the Call method at \$47.59, and the Email + Call method at \$183.63. These baseline values will serve as reference points for future comparisons, allowing the business to measure improvements in its sales strategy over time. Regular monitoring of this metric will enable Pens and Printers to quickly identify trends, adjust strategies based on customer behavior, and optimize resource allocation to the most effective communication methods.

Recommendations

Based on the analysis, the following recommendations are made for optimizing the sales strategy at Pens and Printers:

1. Focus on Email Sales Method: Prioritize the Email communication method, given its proven efficiency in revenue generation and customer engagement.
2. Reassess Call Strategy: Consider reducing resources allocated to the Call method, as its revenue contribution remains significantly lower than Email.
3. Enhance Email Marketing: Leverage insights from customer behavior to tailor email marketing strategies, targeting high-revenue states and emphasizing customer retention.
4. Continuous Monitoring: Implement regular analysis of sales data to adjust strategies based on emerging trends and customer preferences and based on the Revenue Per Customer KPI.

Conclusion

This report provides a comprehensive analysis of customer engagement strategies at Pens and Printers, revealing critical insights into sales performance and customer behavior. By focusing on the most effective communication methods and adjusting resource allocation accordingly, the company can enhance its sales strategy, improve customer engagement, and drive overall business success.

For further information, consult the notebook :

When you have finished...

- Publish your Workspace using the option on the left
- Check the published version of your report:
 - Can you see everything you want us to grade?
 - Are all the graphics visible?
- Review the grading rubric. Have you included everything that will be graded?
- Head back to the [Certification Dashboard](#)  to submit your practical exam report and record your presentation